Nikolay Malkovsky

Principal Researcher | Scientific computing | High-Performance C++

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WORK EXPERIENCE

Principal Researcher 2023–Present
Huawei | 2012 Laboratories

- Improved QUIC for file transfer and real-time communication using congestion-control algorithms.
- Technical lead for R&D on forward error correction in real-time data transfer, including advanced Reed–Solomon and convolutional-code implementations.
- Authored the TechArena 2025 SPbSU challenge (codeforces.com/gym/106111).

Developer 2022–2023
Yandex | Yandex Market

Senior Researcher 2021–2022
TCS Group | AI Department

- Built transformer-based zero-shot text cleaning.
- Developed core components for CTC end-to-end ASR and WFST-based context biasing.

Researcher / Senior Researcher 2018–2021
STC Innovations | Research Department

- Developed the HMM component of hybrid HMM–DNN ASR: finite-automata word-boundary extraction, low-latency streaming recognition, vocabulary expansion, lattice generation, confusion-network conversion, and RNN-LM lattice rescoring.

Assistant Professor 2017–2022
St. Petersburg State University | and HSE University, St. Petersburg campus

- Taught a Russian-language course on convex optimization (course materials).

OPEN SOURCE

Author and Maintainer 2025–Present
Pixie | C++20 succinct data structures library

- Highly optimized compressed data structures with fast access without full decompression, focused on practical memory efficiency and CPU performance.
- github.com/Malkovsky/pixie

Author and Maintainer 2026–Present
AI for C++ | Tools for AI usage in HPC and algorithmic tasks mainly for C++

- Recommended MCPs, skills, general guidance for HPC workflows
- github.com/Malkovsky/ai_for_cpp

Author 2018–2024
Interactive visualization | Python library for interactive widgets

- Set of tools for creating interactive visualization in Python mainly designed for educational purposes
- github.com/Malkovsky/interactive-visualization/

JOB PREFERENCES

I specialize in scientific computing, throughout my career I show best proficiency in projects with high demand on math and high performing algorithms which is also a personal preference in a projects. Ideal project I seek for is the a software engineer position at an R&D department in a fields like NN inference, algebraic computations, big data processing.

EDUCATION

Ph.D. in Computer Science
2013–2017
St. Petersburg State University
St. Petersburg, Russia

Thesis: "*Randomized resource distribution algorithms in multiagent systems.*"
Supervisor: **Prof. Oleg Granichin**.
Candidate of Science (Candidate of Physical and Mathematical Sciences in the Russian degree system).

Master's Degree 2008–2013
St. Petersburg State University
St. Petersburg, Russia
GPA: 4.5/5

LANGUAGES

Russian Native
English Fluent

SKILLS

Algorithms & systems
As a techlead developed performance-critical real-time systems with demand of advanced algorithms: graph and finite-automata algorithms; finite fields and coding theory; succinct data structures.

C / C++
Primary languages; leadership and maintenance of algorithmically rich projects at 10^5–10^6

AI adoptation
I use AI a lot in my workflow including HPC and algorithmic tasks, constantly seek for a ways to improve efficiency of AI tools

Python
Main scripting language: NumPy, SciPy, scikit-learn, Matplotlib, PyTorch, and TorchScript; C/C++ extensions for bottlenecks.

Mathematics
Strong mathematical background applied throughout major research and engineering projects.

Speech recognition
Kaldi, DeepSpeech, NVIDIA NeMo; HMM–DNN, CTC, WFST, context biasing.

Infrastructure
Docker, gRPC, Kubernetes; microservice development.